

01 / THE BASIC IDEA

Forces, potentials and energy

A visual guide to mechanical, electrical, magnetic and thermal behaviour

Two questions to keep separate

Conservative? Can the work be found from an endpoint potential-energy difference?

Dissipative? Does the process turn recoverable energy into internal energy through irreversible changes?

Start with work

A force does work when it has a component along the displacement:

$$W_{A \rightarrow B} = \int_A^B \mathbf{F} \cdot d\mathbf{r} \quad \text{For constant force: } W = Fs \cos \theta.$$

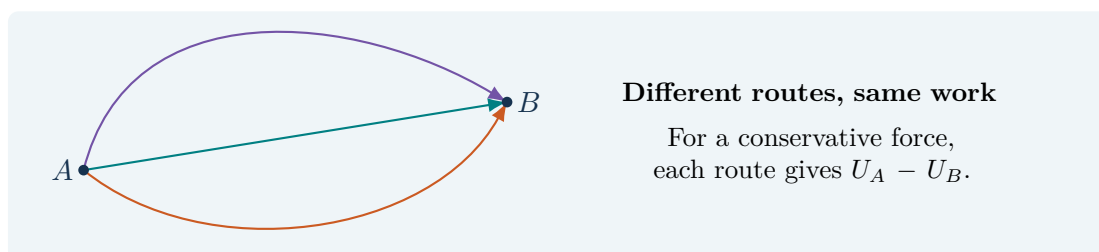
Here s is displacement magnitude and θ is the angle between force and displacement. A perpendicular force does no work.

A conservative force remembers only the endpoints

For an ordinary position-dependent conservative force, there is a potential energy U such that

$$W_{A \rightarrow B} = U_A - U_B = -\Delta U, \quad \mathbf{F} = -\nabla U.$$

In one dimension, $F_x = -dU/dx$: the force points towards decreasing potential energy. The symbol ∇U describes how U changes with position.



For a closed path, the object returns to the same potential energy:

$$\oint \mathbf{F} \cdot d\mathbf{r} = 0.$$

If an ordinary force does nonzero net work around a closed path, it is non-conservative. The usual test assumes a position-only force; magnetic forces on moving charges need special care.

Energy is still conserved

“Non-conservative” does **not** mean energy disappears. Mechanical energy may become internal energy, or energy may be supplied by an external source.

02 / MECHANICAL

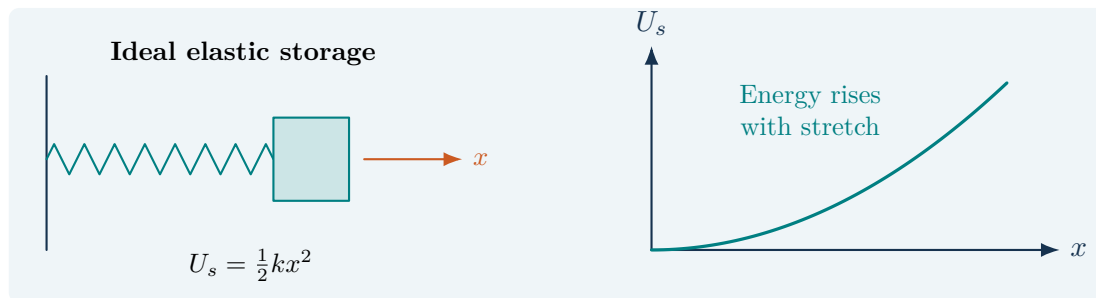
Store it, recover it, or dissipate it

Gravity and an ideal spring are conservative

Near Earth's surface, $U_g = mgh$. Gravity does work $mg(h_A - h_B)$: only the change in height matters. An ideal Hookean spring has

$$F_x = -kx, \quad U_s = \frac{1}{2}kx^2.$$

Stretching stores energy; releasing the spring returns it. This is a simple model for reversible elastic deformation.



Sliding friction is non-conservative and dissipative

For constant kinetic friction magnitude f , sliding a distance L gives

$$W_f = -fL.$$

The minus sign means friction removes mechanical energy. On the return trip friction again opposes motion: the work does not cancel.

Worked example: an out-and-back trip

A block slides 2 m out and 2 m back, with $f = 5$ N.

$$W_f = -5(2 + 2) = -20 \text{ J.}$$

The endpoints are identical, but the work is nonzero. That rules out a position-only potential energy for friction.

Real materials: elastic recovery versus plastic change

During ideal elastic deformation, work is stored and recovered. Plastic deformation changes the internal structure, including defects. Part of the work becomes heat and part can remain stored in the changed structure.

For uniaxial deformation, work per unit volume is $\int \sigma d\varepsilon$, where σ is stress and ε is strain. For a stabilized closed loading cycle that returns the internal state to its starting condition, the loop area is the dissipated energy per unit volume.

Quick check

Does unloading a plastically deformed specimen to zero force guarantee its original shape?

Answer: No. Permanent strain can remain; zero load does not mean the original material state has been restored.

03A / ELECTROSTATIC ENERGY

Where does the electric field enter U ?

Define energy from the work done by the force

Take a charge q in a static external field $\mathbf{E}$ produced by fixed source charges. The external field excludes the charge's own field. For a small displacement,

$$dW_{\text{field}} = \mathbf{F} \cdot d\mathbf{r} = q\mathbf{E} \cdot d\mathbf{r}, \quad dU = -dW_{\text{field}}.$$

Integrating from a reference position $\mathbf{r}_0$ to $\mathbf{r}$ gives

$$U(\mathbf{r}) - U(\mathbf{r}_0) = -q \int_{\mathbf{r}_0}^{\mathbf{r}} \mathbf{E} \cdot d\boldsymbol{\ell}.$$

Thus U is the energy of interaction between the charge and its sources, often called “the charge's potential energy”. Moving the charge slowly with an external agent gives $W_{\text{agent}} = \Delta U$ if no energy is dissipated.

Electric potential packages the field integral

Define $V(\mathbf{r}) - V(\mathbf{r}_0) = -\int_{\mathbf{r}_0}^{\mathbf{r}} \mathbf{E} \cdot d\boldsymbol{\ell}$.

Then $\Delta U = q\Delta V$. With compatible zero references, $U = qV$.

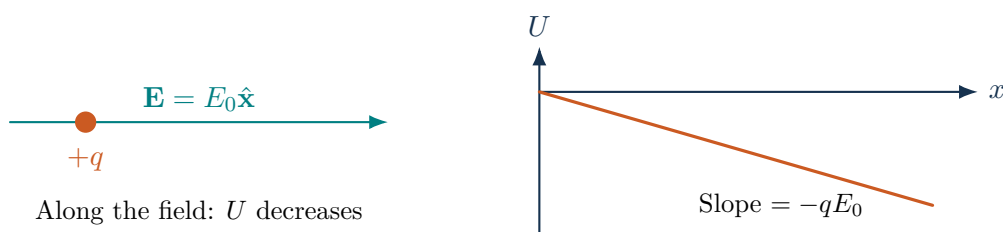
V is measured in volts (J/C); U is measured in joules. Only differences matter. The integral is path independent in electrostatics.

Example 1: a uniform field along $+x$

With $\mathbf{E} = E_0\hat{\mathbf{x}}$ and $U(0) = 0$,

$$U(x) = -qE_0x, \quad -\frac{dU}{dx} = qE_0 = F_x.$$

For $q > 0$, moving along the field lowers U . For $q < 0$, it raises U .

**Example 2: the field of a point source charge Q_s**

In vacuum, $\mathbf{E}(\mathbf{r}) = Q_s\hat{\mathbf{r}}/(4\pi\epsilon_0r^2)$. Choosing $U(\infty) = 0$,

$$U(r) = -q \int_{\infty}^r \frac{Q_s}{4\pi\epsilon_0r'^2} dr' = \boxed{\frac{qQ_s}{4\pi\epsilon_0r}}.$$

Like charges have positive interaction energy; unlike charges have negative interaction energy relative to infinite separation.

A different quantity: energy stored in the field

For an electrostatic field in vacuum, $u_E = \frac{1}{2}\epsilon_0 E^2$ is energy *per unit volume*; total field energy is $\int u_E dV$. This describes the field of the whole charge arrangement, whereas qV_{external} describes a charge's interaction with prescribed sources. They are not interchangeable formulas. Ideal point-charge self-energies diverge; the finite pair interaction above excludes those self-energies. Further reading: OpenStax, 7.1, 7.2 and 8.3.

03B / ELECTRICAL

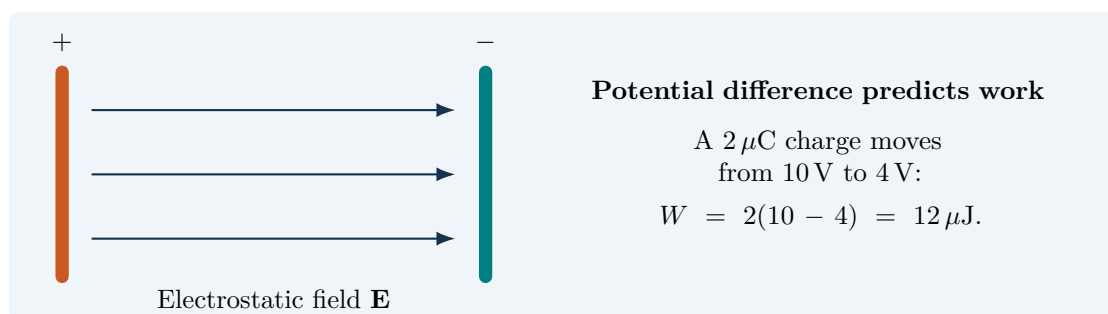
Two kinds of electric field

Electrostatic fields are conservative

For a charge q , $\mathbf{F} = q\mathbf{E}$. Electric potential V is potential energy per unit charge:

$$U = qV, \quad \mathbf{E} = -\nabla V, \quad W_{A \rightarrow B} = q(V_A - V_B).$$

In electrostatics, $\oint \mathbf{E} \cdot d\mathbf{r} = 0$. A positive charge is pushed towards lower V ; a negative charge is pushed in the opposite direction.

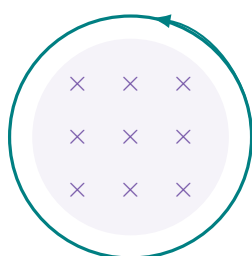


A changing magnetic flux induces a non-conservative field

Magnetic flux $\Phi_B = \int \mathbf{B} \cdot d\mathbf{A}$ measures the magnetic field passing through a surface. For a fixed loop, Faraday's law states

$$\oint \mathbf{E} \cdot d\mathbf{r} = -\frac{d\Phi_B}{dt}.$$

If the linked flux changes, the induced electric field can do net work around the loop. A single-valued scalar potential V alone cannot describe it.



Increasing $\mathbf{B}$ into the page

An electric field can circulate

The induced field points around the loop.
The energy comes from whatever changes
the magnetic field.

Materials connection: resistance and induction heating

Induced electric fields drive currents in metals. Scattering transfers energy from charge carriers to the material, producing Joule heating. **Heating alone does not identify a non-conservative electric field:** resistance also produces heat in a steady DC circuit with an electrostatic field inside the resistor.

04A / MAGNETIC

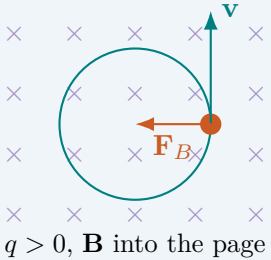
A force can bend motion without doing work

A moving point charge: magnetic force does no work

The magnetic part of the Lorentz force is

$$\boxed{\mathbf{F}_B = q \mathbf{v} \times \mathbf{B}}, \quad \boxed{P_B = \mathbf{F}_B \cdot \mathbf{v} = 0}.$$

The cross product makes the force perpendicular to velocity. Therefore, the magnetic force changes the direction of motion but does not directly change the charge's kinetic energy.



Circular motion in a uniform field

With $\mathbf{v} \perp \mathbf{B}$, magnetic force alone gives circular motion.
Speed stays constant; direction changes.
A changing field may also induce an electric field, which *can* do work.

$q > 0$, $\mathbf{B}$ into the page

Why the usual label needs care

This force depends on velocity, so it is not an ordinary force $-\nabla U(\mathbf{r})$. Yet its work is always zero. The clearest description is a **velocity-dependent, no-work force**. It is not dissipative like friction. The closed-path test for position-only forces should not be applied without this qualification.

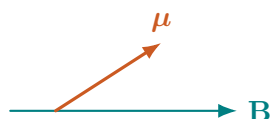
“Magnetic potential” can mean different things

The **vector potential** $\mathbf{A}$ describes the field through $\mathbf{B} = \nabla \times \mathbf{A}$. It is not the scalar potential energy U . Here $\nabla \times$ (curl) describes the local circulation of a vector field; you do not need to calculate it to understand the distinction.

A **fixed magnetic dipole** in a static external field does have interaction potential energy:

$$U = -\boldsymbol{\mu} \cdot \mathbf{B} = -\mu B \cos \theta, \quad \mathbf{F} = \nabla(\boldsymbol{\mu} \cdot \mathbf{B}).$$

The dipole moment $\boldsymbol{\mu}$ describes the strength and orientation of a small magnet. Alignment lowers U . A uniform field provides a torque but no net translational force; a field gradient can pull the dipole.



The dipole tends to align with the field.
This ideal model assumes a fixed moment and no hysteresis.

A dipole has internal structure. Its interaction energy includes the dipole and field sources, so this model does not contradict the point-charge no-work result.

04B / MAGNETIC ENERGY

When is a magnetic interaction conservative?

A small permanent magnet is a dipole, not a point charge

Model a small rigid magnet by a moment μ of constant magnitude, fixed relative to the magnet's body. It may translate and rotate in a prescribed static external field. Assume reversible motion and no magnetic hysteresis or induced-current damping.

Its mechanical interaction can then be described by

$$U(\mathbf{r}, \theta) = -\mu B(\mathbf{r}) \cos \theta = -\boldsymbol{\mu} \cdot \mathbf{B}(\mathbf{r}).$$

Here θ is the angle between the moment and the local field. **Position and orientation both specify the configuration.** Constant moment magnitude does not mean that its direction must stay fixed in space.

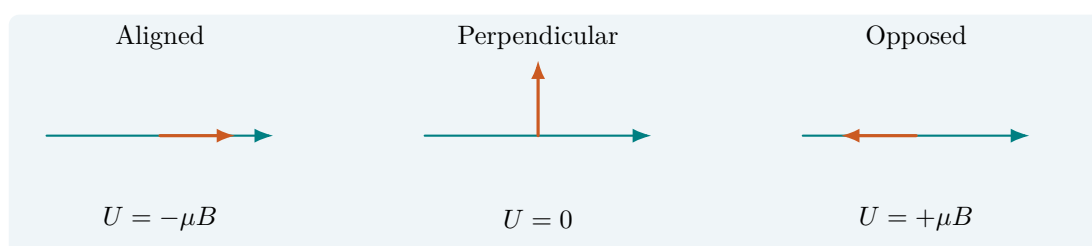
Derive the energy from the torque

The magnetic torque is $\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B}$. For rotation in a fixed plane, taking increasing θ as positive, its signed component is

$$\tau_\theta = -\mu B \sin \theta = -\frac{\partial U}{\partial \theta}.$$

Hence $dU = \mu B \sin \theta d\theta$. At fixed position, integrate from $\theta = \pi/2$, choosing $U(\pi/2) = 0$:

$$U(\theta) = \int_{\pi/2}^{\theta} \mu B \sin \theta' d\theta' = -\mu B \cos \theta.$$



Teal arrows show $\mathbf{B}$; orange arrows show $\boldsymbol{\mu}$. Aligned is a stable energy minimum; exactly opposed is an unstable maximum. Both have zero torque at the exact orientation. Without damping, a displaced dipole can oscillate about alignment; it does not automatically settle there.

Translation: a gradient of field is needed

Holding orientation fixed while taking the spatial derivative,

$$\mathbf{F} = -\nabla_{\mathbf{r}} U = \nabla_{\mathbf{r}} (\boldsymbol{\mu} \cdot \mathbf{B}).$$

A uniform field gives no net translational force on an ideal dipole, but it can exert a torque. Along a path coordinate x , if the dipole is held parallel to a field of constant direction along that path, $F_x = \mu dB/dx$: it is pulled towards stronger field.

A numerical check

For $\mu = 0.020 \text{ A m}^2$, moving aligned from $B_A = 0.10 \text{ T}$ to $B_B = 0.30 \text{ T}$ gives $\Delta U = -\mu(B_B - B_A) = -0.004 \text{ J}$. The interaction does $+0.004 \text{ J}$ of mechanical work. Reversing the same process cancels that work.

Derivation: [OpenStax 11.5](#); force and mechanical energy: [Feynman II, 15–1](#).

04C / ASSUMPTIONS & LIMITS

What must return in a closed cycle?

Include rotation as well as translation

For the ideal permanent dipole in a static field,

$$dW_{\text{mech}} = \mathbf{F} \cdot d\mathbf{r} + \boldsymbol{\tau} \cdot d\boldsymbol{\varphi} = -dU.$$

The small vector $d\boldsymbol{\varphi}$ describes the axis and angle of rotation. Therefore,

$$\oint (\mathbf{F} \cdot d\mathbf{r} + \boldsymbol{\tau} \cdot d\boldsymbol{\varphi}) = 0$$

when both position and orientation return to their initial values. Translational work alone need not cancel if rotational work or a change in orientation is omitted.

Returning to a place is not always returning to a state

A magnet can return to the same location while pointing in another direction. Its potential energy may then differ. This is not evidence of a non-conservative interaction: the full configuration has not returned.

Four cases that must be distinguished

1. **A point charge moving in a magnetic field.** Its nonzero Lorentz force depends on velocity. A position-only potential cannot produce it: reversing velocity reverses the force at the same point. Nevertheless $\mathbf{F}_B \cdot \mathbf{v} = 0$. Calling it “non-conservative” without saying “non-dissipative and no-work” can mislead students.
2. **A permanent dipole in a static prescribed field.** With a fixed moment magnitude and reversible response, $-\boldsymbol{\mu} \cdot \mathbf{B}$ gives a conservative mechanical interaction in position and orientation. Ordinary friction can still dissipate energy without changing this property of the magnetic interaction.
3. **A field that changes with time.** Even if an instantaneous dipole model uses $U(\mathbf{r}, \theta, t) = -\boldsymbol{\mu} \cdot \mathbf{B}(\mathbf{r}, t)$, it is no longer a time-independent conservative system. With no other mechanical forces, $d(K + U)/dt = \partial U/\partial t$. The external field source can exchange energy with it; induced electric fields may also matter.
4. **A real material with changing magnetization.** Hysteresis means the internal magnetic state depends on history. A potential depending only on position and applied field cannot capture a dissipative cycle. A reversible induced moment can admit an effective energy, but its magnetization energy must be included; simply substituting a variable $\mu(B)$ into the permanent-dipole formula is not generally correct.

How can a magnet move if magnetic force does no work?

A point charge and a whole current loop are different systems. Charge velocity includes both the loop’s motion and the motion of charge relative to the loop. Zero total magnetic work on each charge does not imply zero mechanical force on the loop. Energy accounting must include electrical and internal degrees of freedom.

For a loop maintained at fixed current, $-\boldsymbol{\mu} \cdot \mathbf{B}$ is an *effective mechanical potential*; it is not the total field-plus-circuit energy. The power supplies maintaining the currents can exchange energy as the loop moves. This resolves the apparent contradiction without abandoning energy conservation.

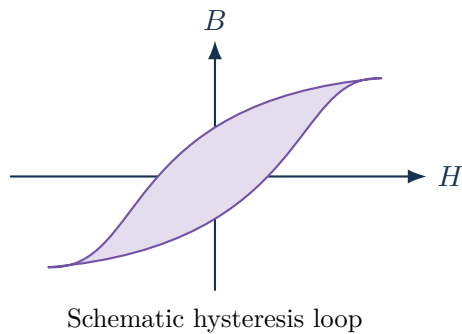
Further explanation of current sources and energy: [Feynman Lectures II, 15–2](#).

05 / THERMAL & MAGNETIC LOSSES

Irreversibility is about the process

Magnetic hysteresis: history matters

In a ferromagnet, domain motion and magnetization changes depend on previous loading. Increasing and decreasing the applied field follow different curves. A complete, repeatable cycle dissipates energy.



Loop area = energy loss per volume

For a quasistatic, collinear, repeatable cycle:

$$w_{\text{loss}} = \oint H dB$$

Units: J m^{-3} .

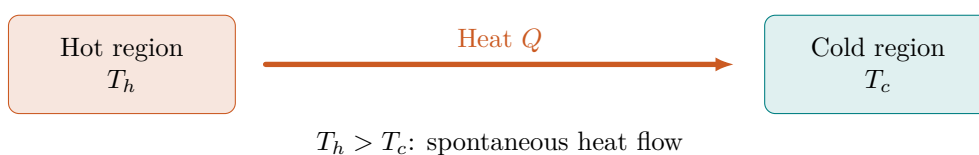
The loss is a property of the material's irreversible response. It does not mean the magnetic force on an individual point charge does work. At finite frequency, measured loops can include additional dynamic losses.

Heat transfer and thermodynamic state properties

Heat Q is energy transferred because of a temperature difference. Internal energy U_{int} is a property of the system's state. Using work done *by* the system as positive, the first law is

$$\Delta U_{\text{int}} = Q - W.$$

For the same initial and final states, ΔU_{int} is fixed, but Q and W can differ between processes. For example, if $\Delta U_{\text{int}} = 100 \text{ J}$, a process with $W = 0$ needs $Q = 100 \text{ J}$, whereas one with $W = 40 \text{ J}$ needs $Q = 140 \text{ J}$.



Heat conduction is irreversible

Fourier's law, $\mathbf{q} = -k\nabla T$, relates heat flux $\mathbf{q}$ to temperature gradient; k is thermal conductivity. For heat $Q > 0$ passed between two reservoirs at fixed absolute temperatures,

$$\Delta S_{\text{total}} = Q \left(\frac{1}{T_c} - \frac{1}{T_h} \right) > 0.$$

Entropy increases while total energy is conserved. Temperature acts as a transport driving potential, not a mechanical potential energy. Sections 15–20 develop enthalpy, free energy, heat-flow gradients and elastic stress. **Call conduction irreversible, rather than calling heat a non-conservative force.**

06 / CONNECT & CHECK

Choose the right description

Example	Description	Reason
Static gravity; ideal spring	Conservative force	Work is an endpoint potential-energy difference.
Sliding friction	Non-conservative, dissipative	Work depends on sliding distance; mechanical energy is degraded.
Plastic deformation	Irreversible material process	Permanent structural change and heating occur.
Electrostatic force	Conservative force	$U = qV$; closed-loop work is zero.
Induced electric field	Non-conservative field	A changing linked flux gives nonzero electric circulation.
Magnetic force on a point charge	Velocity-dependent, no-work force	$\mathbf{F}_B \perp \mathbf{v}$; no direct kinetic-energy change.
Fixed dipole; static external field	Conservative ideal interaction	$U = -\boldsymbol{\mu} \cdot \mathbf{B}$ under the stated assumptions.
Magnetic hysteresis	Dissipative material process	A repeatable magnetic cycle loses recoverable energy.
Heat conduction	Irreversible energy transfer	Heat crosses a temperature difference and generates entropy.

Three common mistakes

- **“Non-conservative means energy is destroyed.”** No. Track the internal energy and external energy transfers as well as mechanical energy.
- **“Anything called a potential is potential energy.”** No. Electric potential is energy per charge; magnetic vector potential is a different mathematical quantity.
- **“Every non-conservative effect is friction.”** No. An induced electric field can supply energy from an external source without itself being a dissipative mechanism.

Quick check

1. A charge completes a circle under a uniform magnetic field alone. What is the magnetic work?
2. An ideal spring is stretched and returned to its initial extension. What is its net work?
3. Is Joule heating sufficient evidence that an electric field is non-conservative?
4. A magnetic material returns to its initial repeatable state after a hysteresis cycle. Was all the supplied energy recovered?

Answers: 1. Zero; force is perpendicular to velocity. 2. Zero; the potential energy returns to its starting value. 3. No; carrier scattering can dissipate energy supplied by an electrostatic field. 4. No; the loop area represents energy loss per unit volume.

The habit to build

Identify the system and assumptions first. Then ask whether you are discussing a **force**, a **material response**, or an **energy transfer**. These are connected, but they are not the same type of quantity.

Notation: U = potential energy; U_{int} = internal energy; V = electric potential; $\mathbf{A}$ = magnetic vector potential; $\mathbf{B}$ = magnetic flux density; H = magnetic field strength. Figures are schematic, not measured data.

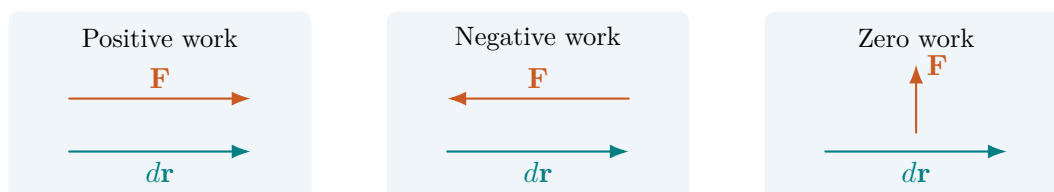
07 / SIGNS FROM FIRST PRINCIPLES

A minus sign needs a sentence

A sign may describe a direction, an energy transfer, or a reference level. These meanings must not be mixed. Throughout this note, a change means **final minus initial**: $\Delta U = U_B - U_A$.

1. Positive work means force helps the displacement

For a small movement, $dW = \mathbf{F} \cdot d\mathbf{r} = F ds \cos \alpha$. Here F and ds are nonnegative magnitudes; α supplies the directional information.



A negative force component does not automatically mean negative work. If both F_x and dx are negative, $F_x dx$ is positive: force and movement both point left.

2. Why is conservative work $W_c = -\Delta U$?

The work–energy theorem states $\Delta K = W_{\text{net}}$. If only a conservative force does work, $\Delta K = W_c$. Conservation of mechanical energy gives

$$\Delta K + \Delta U = 0 \quad \Rightarrow \quad \boxed{W_c = -\Delta U = U_A - U_B}.$$

If the force gives the object 8 J of kinetic energy, the interaction loses 8 J of potential energy. The minus sign expresses that exchange. It does not assert that U itself is negative.

3. Why is work done *against* the force $+\Delta U$?

Suppose an external agent moves the object slowly, with no dissipation and no change in kinetic energy. Then $W_c + W_{\text{agent}} = 0$, so

$$\boxed{W_{\text{agent}} = +\Delta U}, \quad \boxed{W_c = -\Delta U}.$$

These are the works of **two different forces** during the same displacement. If the object accelerates or dissipates energy, do not assume $W_{\text{agent}} = \Delta U$. With only those two forces, $W_{\text{agent}} = \Delta K + \Delta U$.

4. Negative energy is different from decreasing energy

A reference determines the numerical sign of U . Replacing U by $U + C$ leaves both ΔU and $\mathbf{F} = -\nabla U$ unchanged. In contrast, a negative ΔU means a real decrease relative to the initial state.

Read the number before interpreting it

From -10 J to -4 J: $\Delta U = +6$ J; energy *increases*.

From $+10$ J to $+4$ J: $\Delta U = -6$ J; energy *decreases*.

Neither the sign of U nor the sign of work tells you, by itself, whether a force is conservative.

08 / MECHANICAL SIGNS

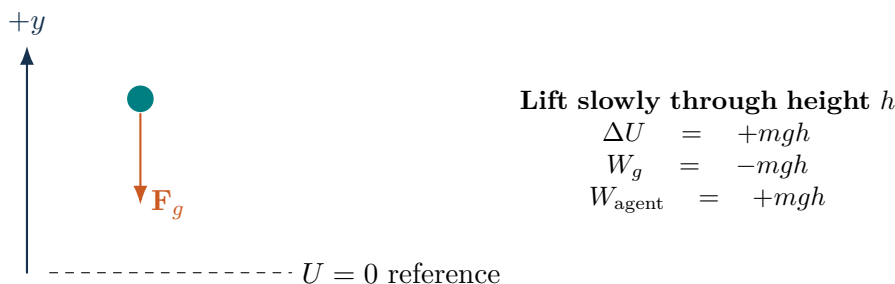
Why U can be positive while force is negative

Gravity: choose the upward direction as positive

Let y increase upwards and set $U_g(0) = 0$. Near Earth's surface,

$$U_g = mgy, \quad F_y = -\frac{dU_g}{dy} = -mg.$$

The energy increases upwards; the force points downwards. The plus sign in mgy and minus sign in $-mg$ agree with each other.



On falling through the same height, all three changes reverse (for the agent, if it lowers the object slowly). Gravity does positive work while U_g decreases. Below the chosen zero height, $U_g < 0$, but gravity still points down.

The spring: the derivative creates the restoring sign

Measure extension x from the unstretched equilibrium position:

$$U_s = \frac{1}{2}kx^2, \quad F_x = -\frac{dU_s}{dx} = -kx.$$

Since $k > 0$ and $x^2 \geq 0$, this choice makes $U_s \geq 0$. But force is signed: for $x > 0$, $F_x < 0$; for $x < 0$, $F_x > 0$. Both point towards equilibrium.

Worked example: stretching and releasing

For $k = 100 \text{ N/m}$, stretch from 0 to 0.10 m.

$$\Delta U_s = \frac{1}{2}(100)(0.10)^2 = +0.50 \text{ J}.$$

The spring does -0.50 J of work; a slow external pull does $+0.50 \text{ J}$. On returning to $x = 0$, the spring does $+0.50 \text{ J}$ and $\Delta U_s = -0.50 \text{ J}$.

Friction: distinguish a signed component from a magnitude

On a stationary surface, kinetic friction on a sliding block opposes its velocity. If $f > 0$ is its magnitude,

$$\mathbf{F}_f = -f \frac{\mathbf{v}}{|\mathbf{v}|}, \quad P_f = \mathbf{F}_f \cdot \mathbf{v} = -f|\mathbf{v}| < 0.$$

Sliding right gives $F_{f,x} = -f$ and $dx > 0$. Sliding left gives $F_{f,x} = +f$ and $dx < 0$. In both cases the work is negative. Therefore $W_f = -fL$ for constant f and total distance $L \geq 0$.

This statement assumes a stationary surface and a sliding block. Friction can do positive work on one object driven by a moving belt; dissipation concerns the relative sliding and the full energy balance.

Quick check

Is the spring non-conservative because it does negative work while stretching?

No. It returns that energy on unloading. Friction on the stationary surface removes mechanical energy on both the outward and return trips.

09 / ELECTROSTATIC SIGNS

Why ΔV has a minus, but $U = qV$ does not**Follow one chain of definitions**

For a constant test charge q in a static external electric field,

$$\begin{aligned}\mathbf{F}_e &= q\mathbf{E}, \\ W_e &= q \int_A^B \mathbf{E} \cdot d\mathbf{r}, \\ \Delta U &= -W_e = -q \int_A^B \mathbf{E} \cdot d\mathbf{r}, \\ \Delta V &\equiv - \int_A^B \mathbf{E} \cdot d\mathbf{r}.\end{aligned}$$

Substitute the last line into the preceding one:

$$\boxed{\Delta U = q\Delta V}, \quad \boxed{W_e = -q\Delta V = q(V_A - V_B)}.$$

The minus sign has not disappeared: it is already inside the definition of ΔV . Adding another minus in $U = -qV$ would count it twice and give the wrong force.

Check the formula by differentiating

With compatible references, $U = qV$. Since q is constant,

$$\mathbf{F}_e = -\nabla U = -q\nabla V = q\mathbf{E}, \quad \mathbf{E} = -\nabla V.$$

The two descriptions produce exactly the same force. The direction of $\mathbf{E}$ is defined using a positive test charge, not an electron.

The charge sign changes the physical result

Take two positions with $V_A = 12\text{ V}$ and $V_B = 5\text{ V}$. The field points towards decreasing potential. For motion from A to B , $\Delta V = -7\text{ V}$.

	$q = +2\text{ }\mu\text{C}$	$q = -2\text{ }\mu\text{C}$
$\Delta U = q\Delta V$	$-14\text{ }\mu\text{J}$	$+14\text{ }\mu\text{J}$
$W_e = -\Delta U$	$+14\text{ }\mu\text{J}$	$-14\text{ }\mu\text{J}$
Slow-agent work W_{agent}	$-14\text{ }\mu\text{J}$	$+14\text{ }\mu\text{J}$

For the positive charge the field helps the motion. To keep it slow, the agent restrains it and receives energy. For the negative charge the field opposes this same displacement, so the agent must supply energy.

Lower electric potential is not always lower potential energy

A positive charge tends to accelerate towards lower V . A negative charge tends to accelerate towards higher V . In both cases, starting from rest under the electrostatic force alone, the motion initially lowers $U = qV$.

Uniform-field shortcut

If $\mathbf{E} = E_0\hat{\mathbf{x}}$, then $\Delta V = -E_0\Delta x$ and $\Delta U = -qE_0\Delta x$. Keep q and Δx signed. Do not insert an extra minus simply because the moving particle is an electron: its negative q already supplies that sign. Reference for the potential convention: [OpenStax, section 7.2](#).

10 / REFERENCE LEVELS

Why Coulomb energy has a plus sign

The integral itself contributes a minus sign

Let a source charge Q_s be at the origin. The radial direction $+r$ points away from it, so $E_r = k_e Q_s / r^2$, where $k_e = 1/(4\pi\epsilon_0) > 0$. Choose $U(\infty) = 0$:

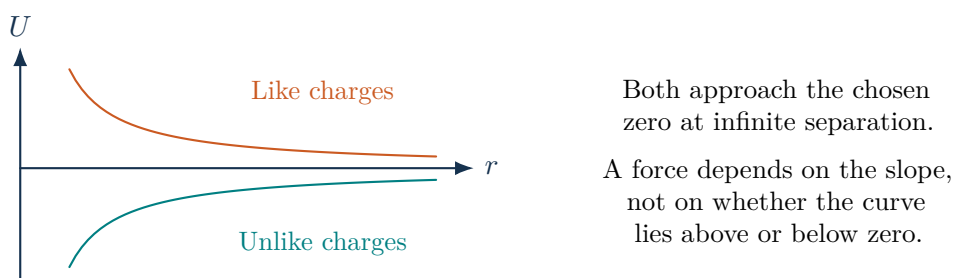
$$\begin{aligned} U(r) &= -q \int_{\infty}^r \frac{k_e Q_s}{r'^2} dr' \\ &= -k_e q Q_s \left[-\frac{1}{r'} \right]_{\infty}^r \\ &= -k_e q Q_s \left(-\frac{1}{r} - 0 \right) = \boxed{+\frac{k_e q Q_s}{r}}. \end{aligned}$$

One minus is from the definition of potential energy. The other comes from $\int r'^{-2} dr' = -1/r'$. Their product is positive. **The product qQ_s can still be negative.**

Read attraction and repulsion from the slope

$$F_r = -\frac{dU}{dr} = -\frac{d}{dr} \left(\frac{k_e q Q_s}{r} \right) = +\frac{k_e q Q_s}{r^2}.$$

For like charges, $qQ_s > 0$: $F_r > 0$, outward repulsion. For unlike charges, $qQ_s < 0$: $F_r < 0$, inward attraction. The correct direction follows without changing the formula.



For unlike charges, $U(r) < 0$ means their interaction energy is below that at infinite separation. Separating them slowly requires positive work $W_{\text{agent}} = 0 - U(r) > 0$. It does not mean their kinetic energy is negative.

Why is electric field energy positive then?

In vacuum, $u_E = \frac{1}{2}\epsilon_0 |\mathbf{E}|^2 \geq 0$ is field energy density. For fields from two finite charge distributions,

$$\frac{\epsilon_0}{2} |\mathbf{E}_1 + \mathbf{E}_2|^2 = \frac{\epsilon_0}{2} E_1^2 + \frac{\epsilon_0}{2} E_2^2 + \epsilon_0 \mathbf{E}_1 \cdot \mathbf{E}_2.$$

The total is nonnegative, but the integrated *interaction term* can be negative. Pair potential energy subtracts the separate self-energies; it is not the entire positive field energy. For ideal point charges those self-energies diverge, so only the finite interaction is used in the Coulomb formula. Coulomb-energy derivation: [OpenStax, section 7.1](#).

11 / MAGNETIC SIGNS

Why $U = -\boldsymbol{\mu} \cdot \mathbf{B}$, but force has a plus

Consider a rigid permanent dipole of constant moment magnitude in a static external field, with reversible magnetic response. Let θ increase as the moment turns *away* from alignment.

Torque magnitude and signed torque are different

For $0 \leq \theta \leq \pi$, the torque magnitude is $|\boldsymbol{\tau}| = \mu B \sin \theta \geq 0$. But the torque acts to *decrease* θ , so its component for the chosen rotation coordinate is

$$\tau_\theta = -\mu B \sin \theta.$$

The vector formula $\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B}$ needs no prefixed minus: the cross product already carries the direction.

Derive, rather than memorize, the energy sign

Rotational work is $dW = \tau_\theta d\theta$ and $dU = -dW$. Thus

$$dU = +\mu B \sin \theta d\theta, \quad U = -\mu B \cos \theta + C.$$

Choose $C = 0$, which sets zero energy at 90° . Then $U = -\boldsymbol{\mu} \cdot \mathbf{B}$. The minus makes alignment the lowest-energy orientation, consistent with the torque.

Worked rotation: keep every sign visible

Let $\mu = 0.10 \text{ A m}^2$ and $B = 0.20 \text{ T}$, so $\mu B = 0.020 \text{ J}$. Rotate from 90° to 0° :

$U_i = 0$, $U_f = -0.020 \text{ J}$, $\Delta U = -0.020 \text{ J}$.

The magnetic torque does $W_{\text{mag}} = +0.020 \text{ J}$. A slow restraining agent does -0.020 J . Turning it back requires the agent to supply $+0.020 \text{ J}$.

The force formula contains two minus signs

At fixed orientation, differentiate with respect to position:

$$\mathbf{F} = -\nabla U = -\nabla(-\boldsymbol{\mu} \cdot \mathbf{B}) = +\nabla(\boldsymbol{\mu} \cdot \mathbf{B}).$$

The plus is the result of the two minuses. It does not mean the force always points in the positive coordinate direction. That depends on the gradient.

Along a path with fixed field direction, a held aligned dipole has $U = -\mu B(x)$ and $F_x = +\mu dB/dx$. If B increases along $+x$, the force is positive. A held opposed dipole has $U = +\mu B(x)$ and $F_x = -\mu dB/dx$. Its force is negative for the same gradient. Without a constraint, the opposed orientation is unstable to rotation.

Quick check

If the field is uniform, can the energy still change?

Yes, by rotation. There is no positional gradient and no net translational force, but $U = -\mu B \cos \theta$ changes with orientation. Translational force and rotational torque are different derivatives of the same energy.

Torque and dipole energy: [OpenStax, section 11.5](#).

12 / WHAT ENERGY ARE WE COUNTING?

Positive field energy and negative dipole energy

They refer to different energy accounts

Magnetic field energy density in vacuum is

$$u_B = \frac{B^2}{2\mu_0} \geq 0.$$

It counts energy in the magnetic field. By contrast, $U_{\text{mech}} = -\boldsymbol{\mu} \cdot \mathbf{B}_{\text{external}}$ predicts the mechanical force and torque on a fixed-moment dipole in a prescribed field. Its zero and system boundary differ from those of the total field energy.

For a loop held at fixed current, motion induces electrical effects. The supplies maintaining the currents can give or receive energy. Differentiating just the field energy while ignoring these exchanges can give an incorrect mechanical force or sign.

Always identify the system

“Energy of the dipole in a prescribed field” and “energy of the field, circuits, dipole and power supplies together” are not the same quantity. A conservative effective mechanical potential does not eliminate electrical energy exchange with the supplies.

An induced moment: why another factor can appear

An ideal permanent moment is already present before applying the field. An induced moment grows as the field is applied. For a simple reversible, collinear model, suppose $\mu_{\text{ind}} = \alpha B$, with constant α . Using $F_x = \mu_{\text{ind}}(B) dB/dx$ for this dipole model gives

$$dU_{\text{eff}} = -F_x dx = -\mu_{\text{ind}}(B) dB,$$

so, with $U_{\text{eff}}(0) = 0$,

$$U_{\text{eff}}(B) = - \int_0^B \mu_{\text{ind}}(b) db = -\frac{1}{2}\alpha B^2.$$

The integration variable b runs through the field values. The factor 1/2 appears because the moment rises from zero to its final value; inserting the final moment into $-\mu B$ throughout would count too much.

For $\alpha > 0$, $F_x = \alpha B dB/dx$ points towards increasing field magnitude. For $\alpha < 0$, it points towards decreasing field magnitude. This is an ideal effective model, not a general formula for hysteretic ferromagnets or arbitrary specimen geometries.

The point-charge result still holds

$$\mathbf{F}_B = q\mathbf{v} \times \mathbf{B}, \quad \mathbf{F}_B \cdot \mathbf{v} = 0.$$

Changing the sign of q reverses the force direction, but the power remains zero. The force does not become dissipative for a negative charge. For a whole magnet or circuit, internal motion and electrical energy exchange must also be counted.

Quick check

Does $U_{\text{mech}} < 0$ prove that the total energy of a magnet and its field is negative?

No. It is an interaction energy with a chosen reference, or an effective mechanical potential under specified external controls.

Further reading: [Feynman II, 15](#); [University of Pisa: induced-dipole energy](#).

13 / THERMAL SIGNS

Why some books write $Q - W$ and others $Q + W$

Both can be correct: the definition of work differs

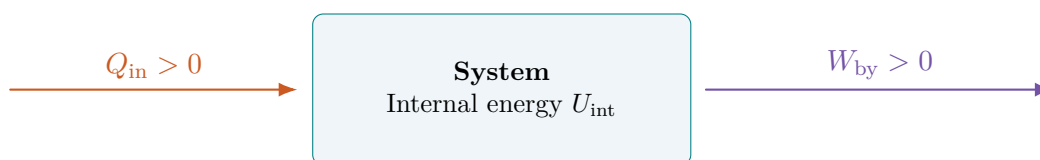
For a closed system, neglecting changes in bulk kinetic and gravitational potential energy, let $Q_{\text{in}} > 0$ mean net heat entering it. If $W_{\text{by}} > 0$ means work done *by* the system on its surroundings,

$$\Delta U_{\text{int}} = Q_{\text{in}} - W_{\text{by}}.$$

Heat entering adds energy. Work delivered to the surroundings removes energy. If instead $W_{\text{on}} > 0$ means work done *on* the system, $W_{\text{on}} = -W_{\text{by}}$ and

$$\Delta U_{\text{int}} = Q_{\text{in}} + W_{\text{on}}.$$

The plus or minus follows the work convention. These equations predict identical changes when used consistently. Here U_{int} is internal energy, not just a mechanical potential energy.



Use one physical example with both conventions

A system receives 100 J of heat and does 30 J of expansion work:

$$\Delta U_{\text{int}} = 100 - 30 = 70 \text{ J} = 100 + (-30) \text{ J}.$$

In the first expression $W_{\text{by}} = +30 \text{ J}$; in the second $W_{\text{on}} = -30 \text{ J}$. During an adiabatic compression with 30 J of work input, $Q = 0$, $W_{\text{on}} = +30 \text{ J}$ and $\Delta U_{\text{int}} = +30 \text{ J}$.

Fourier's minus sign gives the direction of heat flow

In one dimension,

$$q_x = -k \frac{dT}{dx}, \quad k > 0.$$

If temperature increases to the right, $dT/dx > 0$ and $q_x < 0$: heat flows left, towards colder material. If the right side is colder, $dT/dx < 0$ and $q_x > 0$: heat flows right. Heat flux is a signed component, not a negative quantity of stored heat.

Entropy signs: a part can lose entropy while the total gains

Let $Q > 0$ pass from a hot reservoir at fixed T_h to a cold reservoir at fixed $T_c < T_h$. Temperatures are in kelvin:

$$\Delta S_h = -\frac{Q}{T_h}, \quad \Delta S_c = +\frac{Q}{T_c}, \quad \Delta S_{\text{total}} = Q \left(\frac{1}{T_c} - \frac{1}{T_h} \right) > 0.$$

The hot reservoir loses entropy; the cold reservoir gains more than that loss. The positive total expresses irreversibility. It does not mean that every part of the system must have a positive entropy change.

Work convention: [OpenStax, section 3.3](#).

14 / INDUCTION & A SIGN-CHECK ROUTINE

What Faraday's minus sign actually opposes

First choose a loop direction and its matching normal

For a fixed loop, choose a positive direction around it. Curl the fingers of your right hand in that direction; the thumb gives the positive surface normal. Define signed magnetic flux with that normal:

$$\Phi_B = \int \mathbf{B} \cdot d\mathbf{A}, \quad \mathcal{E} = \oint \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi_B}{dt}.$$

A negative $\mathcal{E}$ means circulation opposite to your chosen loop direction. It does not mean energy is necessarily being lost. See [OpenStax 13.2](#).

Lenz's law opposes the *change*, not always the field

Choose counterclockwise as positive when viewing the page, so the positive normal points out of the page. For a simple conducting loop:

Applied flux	Sign of induced emf	Induced response
Out of page, increasing	$d\Phi_B/dt > 0, \mathcal{E} < 0$	Clockwise current; induced field into page
Out of page, decreasing	$d\Phi_B/dt < 0, \mathcal{E} > 0$	Counterclockwise current; induced field out of page

In the second case the induced field supports the original field because it resists its *decrease*. With appreciable self-inductance, distinguish the induced emf from the current's subsequent time evolution.

The minus signs do not all have the same origin

- $W_c = -\Delta U$: mechanical work by a conservative interaction is supplied by decreasing potential energy.
- $\mathbf{E} = -\nabla V$ and $\mathbf{F} = -\nabla U$: field or force points down the corresponding scalar gradient.
- $U = -\boldsymbol{\mu} \cdot \mathbf{B}$: dipole alignment lowers the chosen mechanical potential.
- $q_x = -k dT/dx$: heat flows towards lower temperature.
- $\mathcal{E} = -d\Phi_B/dt$: the induced response opposes a flux change with the stated orientation convention.

A minus sign alone does not establish dissipation or show that a force is non-conservative.

A reliable five-step check for any problem

1. **Name the quantity:** force component, work, potential energy, heat or flux?
2. **State the convention:** which direction is positive; work by whom?
3. **Write final minus initial:** never silently reverse the endpoints.
4. **Substitute signed quantities:** q , displacement and gradients already carry signs.
5. **Check the physics:** does the force help the motion; does the energy rise or fall? Differentiate a proposed U to recover the force.

Quick check

Can a conservative force do negative work? **Yes.**
 Can potential energy be negative? **Yes; specify its reference.**
 Does a positive force component guarantee positive work? **No; check displacement.**
 Does a plus in $\nabla(\boldsymbol{\mu} \cdot \mathbf{B})$ contradict $\mathbf{F} = -\nabla U$? **No; U already contains a minus.**

15 / THERMODYNAMIC POTENTIALS

Enthalpy belongs in the energy picture

A thermodynamic potential is a state function

A mechanical potential $U(\mathbf{r})$ predicts a force from position. A thermodynamic potential describes a material's equilibrium state in variables such as entropy, temperature, volume and pressure. It can predict equilibrium and the direction of an allowed change. It is not generally a spatial force potential.

Use $\mathcal{V}$ for volume here, to distinguish it from electric potential V . For a closed, simple compressible system of fixed composition, with only pressure–volume work, the fundamental relation between equilibrium states is

$$dU_{\text{int}} = T dS - p d\mathcal{V}.$$

Here T is absolute temperature, S entropy and p pressure. Expansion does positive work on the surroundings, explaining the minus before $p d\mathcal{V}$. The relation identifies

$$T = \left(\frac{\partial U_{\text{int}}}{\partial S} \right)_{\mathcal{V}}, \quad p = - \left(\frac{\partial U_{\text{int}}}{\partial \mathcal{V}} \right)_S.$$

Temperature and pressure are derivatives of a thermodynamic potential with respect to state variables. They are not spatial gradients of energy.

Why introduce $H = U_{\text{int}} + p\mathcal{V}$?

At controlled pressure, expansion work accompanies heating or a phase change. Enthalpy combines internal energy with the pressure–volume term:

$$\begin{aligned} H &= U_{\text{int}} + p\mathcal{V}, \\ dH &= dU_{\text{int}} + p d\mathcal{V} + \mathcal{V} dp \\ &= T dS + \underbrace{-p d\mathcal{V} + p d\mathcal{V}}_0 + \mathcal{V} dp = T dS + \mathcal{V} dp. \end{aligned}$$

The added term removes volume as a natural independent variable and introduces pressure. Enthalpy naturally uses (S, p) ; in practice it is often tabulated as $H(T, p)$ through an equation of state.

When does heat equal an enthalpy change?

For a closed system at constant pressure, with only pressure–volume work against that pressure and negligible bulk kinetic and potential energy changes,

$$Q_p = \Delta U_{\text{int}} + p\Delta\mathcal{V} = \Delta H.$$

This is a conditional equality. Enthalpy is a state property; heat is a process-dependent energy transfer. Calling H “heat stored in the body” is misleading.

Materials example: heating and melting

At constant pressure, heating a single phase requires $Q_p = \Delta H = \int C_p dT$. Here C_p is the heat capacity of the whole specimen. During equilibrium melting, heat can enter at nearly constant temperature: it supplies the latent enthalpy of fusion rather than simply raising T .

An exothermic constant-pressure change has $\Delta H < 0$ and releases heat under these assumptions. That sign alone does not establish spontaneity: entropy must also be considered.

Quick check

Is enthalpy the universal potential that always decreases?

No. The relevant equilibrium potential depends on the controlled variables. At constant temperature and pressure, it is Gibbs free energy that provides the usual criterion.

16 / CHOOSING THE POTENTIAL

Why materials engineers use free energy

The surroundings determine the useful potential

For the same simple closed system, define Helmholtz free energy $\mathcal{F}$ and Gibbs free energy G :

$$\mathcal{F} = U_{\text{int}} - TS, \quad G = H - TS = U_{\text{int}} + p\mathcal{V} - TS.$$

The minus TS accounts for the entropy contribution when temperature is controlled. Differentiate, including both terms in $d(TS) = T dS + S dT$:

$$\boxed{d\mathcal{F} = -S dT - p d\mathcal{V}}, \quad \boxed{dG = -S dT + \mathcal{V} dp}.$$

These differential forms describe equilibrium states at fixed composition and with no additional work modes. Allowing composition, strain or other internal variables introduces further terms.

Potential	Natural variables	Equilibrium principle
U_{int}	$S, \mathcal{V}$	Minimum at fixed $S, \mathcal{V}$
$H = U_{\text{int}} + p\mathcal{V}$	S, p	Minimum at fixed S, p
$\mathcal{F} = U_{\text{int}} - TS$	$T, \mathcal{V}$	Minimum at fixed $T, \mathcal{V}$
$G = H - TS$	T, p	Minimum at fixed T, p

The minimum is over allowed internal changes with the stated constraints and conserved amounts held fixed. Stable equilibrium is a minimum; barriers can trap a material in a metastable local minimum. For an isolated system with fixed energy and volume, the equivalent familiar criterion is maximum entropy.

Phase changes: enthalpy and entropy compete

At a common fixed T and p , comparing two material states gives

$$\boxed{\Delta G = \Delta H - T\Delta S}.$$

For melting, both the enthalpy and entropy of fusion are usually positive. Below the equilibrium melting temperature the enthalpy cost dominates; above it the entropy term can make the liquid favourable. At coexistence, $\Delta G = 0$.

Illustrative calculation, per mole

Take $\Delta H_{\text{fus}} = 10 \text{ kJ/mol}$ and $\Delta S_{\text{fus}} = 10 \text{ J/(mol K)}$, treating them as constant over this example.

At 800 K: $\Delta G = 10000 - 800(10) = +2000 \text{ J/mol}$.

At 1200 K: $\Delta G = 10000 - 1200(10) = -2000 \text{ J/mol}$.

The same positive melting enthalpy can accompany different signs of ΔG .

A free-energy driving force is not automatically a conservative force

If ξ measures transformation progress, an appropriate driving force is $X_\xi = -\partial G/\partial \xi$ at fixed external conditions. A dissipative kinetic law can have $\dot{\xi} = LX_\xi$ with $L > 0$, giving $\dot{G} = -LX_\xi^2 \leq 0$.

There is a potential, but the dynamics dissipate available energy. Thus **the existence of a potential does not imply that every process it drives is reversible or mechanically conservative**.

Further reading: [MIT, Thermodynamics of Materials: free-energy criteria](#).

17 / THERMAL DRIVING POTENTIAL

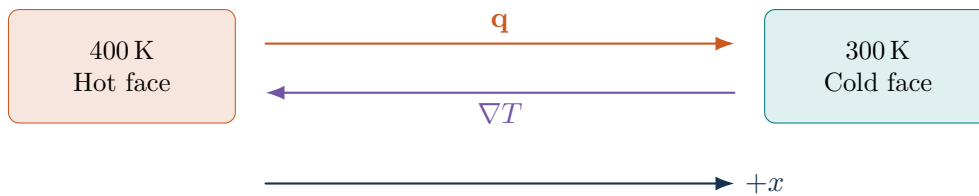
Temperature drives heat flow; entropy tests its loss

Is temperature a thermal potential? In a transport sense, yes

In heat-transfer language, temperature is often treated as a thermal driving potential: a difference in temperature drives heat transfer, just as a voltage difference can drive an electric current. Its spatial gradient is a *driving field*, not an energy stored per unit mass or volume.

$$\mathbf{q} = -k\nabla T, \quad k > 0.$$

$\mathbf{q}$ is heat flux in W/m^2 , ∇T has units K/m , and k has units $\text{W}/(\text{m K})$. The output is a flux, not a mechanical force in newtons.



For a 0.10 m slab with a linear profile, $dT/dx = (300 - 400)/0.10 = -1000 \text{ K}/\text{m}$. If $k = 20 \text{ W}/(\text{m K})$, then $q_x = +20000 \text{ W}/\text{m}^2$, towards the cold face.

The thermodynamic force for heat conduction

For a stationary, simple conductor in local equilibrium, with heat conduction as the only dissipative process, the entropy-production density is

$$\dot{s}_{\text{gen}} = \mathbf{q} \cdot \nabla \left(\frac{1}{T} \right).$$

It has units $\text{W}/(\text{m}^3 \text{ K})$. This formulation calls $\nabla(1/T)$ the thermodynamic force conjugate to heat flux. It is *not* an ordinary mechanical force. Since

$$\nabla(1/T) = -\frac{\nabla T}{T^2}, \quad \mathbf{q} = kT^2 \nabla(1/T),$$

Fourier conduction gives

$$\dot{s}_{\text{gen}} = \frac{k}{T^2} |\nabla T|^2 \geq 0.$$

The minus sign in Fourier's law makes heat flow down the temperature gradient; the equivalent plus sign with $\nabla(1/T)$ gives the same flux. Entropy production is positive whenever the gradient is nonzero.

Why "it is a gradient" does not prove conservative behaviour

For a smooth, single-valued temperature field at a fixed time, $\oint \nabla T \cdot d\mathbf{r} = 0$. But $\nabla T \cdot d\mathbf{r}$ is a temperature difference, not mechanical work. Heat conduction can therefore have a gradient driving field and still be irreversible.

Transport interpretation: [Entropy and the Second Law: the nonequilibrium perspective](#).

18 / ELASTIC POTENTIAL

Stress is a derivative of stored energy

Start with a tensile bar

Take a uniform bar of original length L_0 and area A_0 , at small strain. Its extension is $\delta = L - L_0$, strain is $\varepsilon = \delta/L_0$, and tensile stress is $\sigma = F_{\text{applied}}/A_0$. The original volume is $\mathcal{V}_0 = A_0 L_0$.

$$dW_{\text{on}} = F_{\text{applied}} d\delta = (\sigma A_0)(L_0 d\varepsilon) = \mathcal{V}_0 \sigma d\varepsilon.$$

For a reversible purely elastic model at fixed temperature, define stored elastic energy density ψ_{el} (elastic Helmholtz energy per reference volume). Then

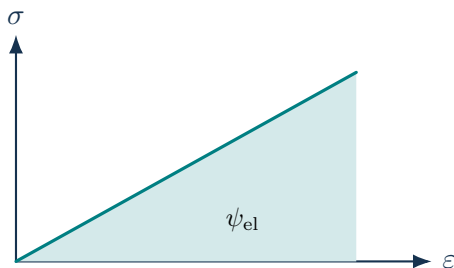
$$\boxed{d\psi_{\text{el}} = \sigma d\varepsilon}, \quad \boxed{\sigma = \frac{\partial \psi_{\text{el}}}{\partial \varepsilon}}.$$

Strain energy density is the elastic potential. Stress is its conjugate response, meaning that stress multiplied by a strain increment gives work per unit volume.

Linear elasticity gives the familiar triangle

Using Young's modulus E_Y (not electric field $\mathbf{E}$), $\sigma = E_Y \varepsilon$. With zero energy at zero strain,

$$\psi_{\text{el}}(\varepsilon) = \int_0^\varepsilon E_Y \varepsilon' d\varepsilon' = \boxed{\frac{1}{2} E_Y \varepsilon^2 = \frac{1}{2} \sigma \varepsilon}.$$



Area under the stress-strain curve
is elastic energy per volume.

The same curve is retraced on unloading
in the ideal reversible model.

Why the plus in stress does not contradict a restoring force

The bar's total stored energy is $U_{\text{el}} = \mathcal{V}_0 \psi_{\text{el}}$. Its restoring force on the moving end is

$$F_{\text{restoring}} = -\frac{dU_{\text{el}}}{d\delta} = -A_0 \sigma.$$

The slow applied force is $+A_0 \sigma$. Stress is positive in tension by convention; the bar pulls its extended end back. The plus derivative with respect to *strain* and minus derivative for the *restoring force* describe different quantities.

Units and a numerical example

1 Pa = 1 J/m³, but stress and energy density still have different physical meanings. For $E_Y = 200$ GPa and $\varepsilon = 0.001$, $\sigma = 200$ MPa and $\psi_{\text{el}} = 100$ kJ/m³. A 1 cm³ specimen stores 0.10 J in this ideal model.

Stress work and conjugate variables: [University of Alberta](#), [deformation energy](#).

19 / THERMOELASTIC MATERIALS

One free energy can connect stress and temperature

Temperature and strain can be independent state variables

For a small-strain, reversible thermoelastic solid of fixed composition, use Helmholtz free-energy density $\psi(T, \epsilon)$ per reference volume. Its differential is

$$d\psi = -s dT + \boldsymbol{\sigma} : d\boldsymbol{\epsilon}.$$

Here s is entropy per reference volume; the colon means summing the work contributions from the stress and strain components. Therefore,

$$s = - \left(\frac{\partial \psi}{\partial T} \right)_{\epsilon}, \quad d\psi|_T = \boldsymbol{\sigma} : d\boldsymbol{\epsilon}.$$

The latter identifies stress as the derivative of ψ with respect to strain. In three dimensions stress is a tensor, so it cannot generally be a single scalar elastic potential. Use compatible tensor or engineering-shear conventions when differentiating components.

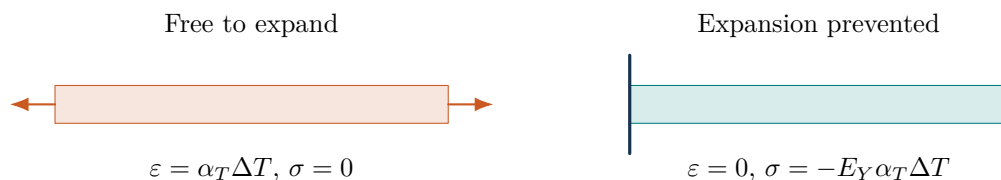
A one-dimensional model explains thermal stress

Let α_T be a constant thermal-expansion coefficient and T_0 a stress-free reference temperature. A simple model is

$$\psi(T, \epsilon) = \psi_{\text{th}}(T) + \frac{1}{2} E_Y [\epsilon - \alpha_T (T - T_0)]^2.$$

The elastic strain is total strain minus free thermal strain. Differentiation at fixed T gives

$$\sigma = E_Y [\epsilon - \alpha_T (T - T_0)].$$



For heating with $\alpha_T > 0$, a restrained bar has negative stress: compression. Taking $E_Y = 200$ GPa, $\alpha_T = 12 \times 10^{-6} \text{ K}^{-1}$ and $\Delta T = 50$ K gives $\sigma = -120$ MPa, provided the response remains elastic.

Pressure is the compressive part of stress

With tensile stress positive, hydrostatic compression has $\boldsymbol{\sigma} = -p\mathbf{I}$. Its small-strain work contribution is $\boldsymbol{\sigma} : d\boldsymbol{\epsilon} = -p \text{tr}(d\boldsymbol{\epsilon})$. Multiplication by reference volume gives approximately $-p dV$, matching the thermodynamic sign.

When does elastic storage stop describing the whole process?

Plastic strain, viscosity, damage or magnetic hysteresis require extra state variables and dissipative laws. A material can retain a recoverable free-energy part while also dissipating energy. In an ideal thermoelastic solid the local constitutive response is reversible; heat conduction between regions can still generate entropy.

The potential and the process answer different questions

$\psi(T, \epsilon)$ describes stored free energy and equilibrium responses. A heat-flow law or a plastic-flow law describes how the material evolves. A well-defined free energy does not remove dissipative mechanisms.

20 / PUTTING THE POTENTIALS TOGETHER

Use the analogy, but keep the quantities distinct

Setting	Potential or energy	Derivative / response
Particle mechanics	Potential energy $U(\mathbf{r})$	$\mathbf{F} = -\nabla_{\mathbf{r}}U$
Electrostatics	Electric potential V ; charge energy $U = qV$	$\mathbf{E} = -\nabla V$; $\mathbf{F} = q\mathbf{E}$
Ideal elasticity	Stored energy density $\psi_{\text{el}}(\epsilon)$	$\sigma = \partial\psi_{\text{el}}/\partial\epsilon$
Thermal and elastic	Free-energy density $\psi(T, \epsilon)$	$s = -\partial\psi/\partial T$; strain derivative gives stress
Simple equilibrium system	$U_{\text{int}}(S, \mathcal{V})$	$T = \partial U_{\text{int}}/\partial S$; $p = -\partial U_{\text{int}}/\partial \mathcal{V}$
Constant-pressure heating	Enthalpy $H = U_{\text{int}} + p\mathcal{V}$	$Q_p = \Delta H$ under the stated work restrictions
Constant- T, p equilibrium	Gibbs energy $G = H - TS$	Allowed spontaneous changes lower G under the stated constraints
Heat conduction	T as transport potential; $1/T$ in entropy formulation	$\mathbf{q} = -k\nabla T$; equivalently $\mathbf{q} = kT^2\nabla(1/T)$
Fixed magnetic dipole	$U_{\text{mech}} = -\boldsymbol{\mu} \cdot \mathbf{B}$	$\mathbf{F} = -\nabla_{\mathbf{r}}U$; torque from orientation derivative

Three meanings of “potential”

1. **A mechanical potential energy** gives work and force for the chosen mechanical coordinates.
2. **A thermodynamic potential** is a state function suited to particular constraints. Its derivatives give conjugate quantities, and its extrema determine equilibrium.
3. **A transport driving potential** varies through space and drives a flux through a transport law. That flux can be dissipative even though the driving field is a gradient.

What to call each quantity

Enthalpy is a thermodynamic potential useful for heat accounting at constant pressure. **Temperature gradients** drive heat conduction, whose irreversibility follows from entropy production. **Elastic stress** is obtained from an elastic energy potential; stress itself is not that stored energy.

Quick check

A. Is σ an elastic potential because its units equal energy per volume?

No. Units alone do not establish the relationship; $\sigma = \partial\psi_{\text{el}}/\partial\epsilon$.

B. Does the existence of H make heat transfer conservative?

No. A state function can describe energy changes during an irreversible process.

C. Why can $\sigma = +\partial\psi/\partial\epsilon$ and $\mathbf{F} = -\nabla U$ both be right?

They use different coordinates and work definitions: stress work supplies stored energy, while the restoring force releases it.